

YEAR 7 · MATHEMATICS

Q1 Review: Units 1 to 3

Mathematics 1.1–1.6 — integers, factors, multiples, tests, roots

Mathematics 2.1–2.6 — expressions, formulae, brackets, equations

Mathematics 3.1–3.2 — powers of 10, place value, rounding

Name: _____

Date given: _____

How to use this packet

Focus the basics · **Practice** like the assessment · **Challenge** the hardest — do not skip
No calculator. Show all your work.

Section A · Unit 1 — Integers, Factors, Multiples and Roots

A1 All four rules, all four signs

FOCUS

Two sets of rules — do not mix them up

Add a negative or subtract a positive: move **left**. Subtract a negative: move **right**.
Two negatives make a positive only for $\times$ and $\div$. $-6 + (-4) = -10$, not $+10$.

(a) $-8 + 5 =$ _____

(d) $-7 \times (-8) =$ _____

(g) $-15 + (-25) =$ _____

(b) $7 + (-12) =$ _____

(e) $54 \div (-6) =$ _____

(h) $0 - (-13) =$ _____

(c) $-3 - (-11) =$ _____

(f) $-81 \div (-9) =$ _____

(i) $-2 \times (-3) \times (-5) =$ _____

A2 Say it, then write it

PRACTICE

Word help — these words decide the calculation

subtract 9 from 4 · **6 less than** -3 start at the **second** number: $4 - 9$ and $-3 - 6$.

by how many degrees

it wants the **change**: the gap between the two temperatures. A gap is never negative.

Write the calculation, then the answer.

1. Subtract 9 from 4.

Calculation: _____ Answer: _____

2. What is 6 less than -3 ? Calculation: _____ Answer: _____

3. At midnight in Moscow the temperature is -11°C . At noon it is 2°C . By how many degrees has the temperature risen?

Calculation: _____ Answer: _____

A3 The missing integer

PRACTICE

Write the missing integer in each box. Check it by putting it back in.

(a) $\square \times (-4) = 36$

(c) $\square \div (-5) = 7$

(e) $\square - 8 = -3$

(b) $-48 \div \square = -6$

(d) $6 \times \square = -42$

(f) $-10 + \square = -25$

A4 The LCM and the HCF, from the lists

FOCUS

One list stops, the other never does

Multiples never stop: we want the **lowest** common one, the **LCM**. **Factors** stop: we want the **highest**, the **HCF**.

(a) Multiples of 8: _____ Multiples of 14: _____

The LCM of 8 and 14 is _____

(b) Factors of 36: _____ Factors of 54: _____

The HCF of 36 and 54 is _____

(c) Find the LCM of 9 and 15. _____ (d) Find the HCF of 28 and 42. _____

(e) Draw a ring around **each** number that is a common factor of 48 and 72.

3 8 9 16 24 36

A5 Which one does the question want?

PRACTICE

Word help – the words that tell you which one

together again · **at the same time** · **the same number of each** – something **repeating**. You want the **LCM**.

equal groups · **as large as possible** · **none left over** · **none wasted** – something being **split up**. You want the **HCF**.

Tick HCF or LCM, then answer.	HCF	LCM	Answer
One light flashes every 4 seconds and another every 10 seconds. They flash together now. After how many seconds will they next flash together?			
Mr Bowen has 30 bananas and 45 mangoes. He makes identical fruit bags, using all the fruit. What is the greatest number of bags he can make?			
Sausages come in packs of 6 and bread rolls in packs of 8. Mr Bowen wants the same number of each. What is the smallest number of sausages he can buy?			

A6 Tests for divisibility

PRACTICE

(a) Tick (✓) where the number divides exactly. Use the tests – do **not** do the divisions.

Number	2	3	4	5	6	9	10
4356							
7290							
3015							

(b) The number 4 8 is divisible by 3. Find **all** the possible missing digits.

(c) Show that 62 417 is **not** divisible by 3. Do not divide.

A7 Squares, cubes and roots

FOCUS

Rooting is squaring run backwards

$9^2 = 81$, so $\sqrt{81} = 9$. $4^3 = 64$, so $\sqrt[3]{64} = 4$. 3^3 is $3 \times 3 \times 3 = 27$, **not** 3×3 .

(a) $13^2 =$ _____

(c) $\sqrt{121} =$ _____

(e) $3^3 - 4^2 =$ _____

(b) $5^3 =$ _____

(d) $\sqrt[3]{1000} =$ _____

(f) $\sqrt{36} + \sqrt[3]{8} =$ _____

(g) Explain why 25 is a square number but **not** a cube number.

A8 Unit 1 puzzles

CHALLENGE

Word help

product — the answer to a **multiplication**. **sum** — the answer to an **addition**.

1. The HCF of two numbers is 5. The LCM of the two numbers is 60. Both numbers are bigger than 5 and smaller than 60. Find the two numbers.

2. The product of two integers is -42 . Their sum is 1. Find the two integers.

3. The number $2 \square 5$ is divisible by 15. Find **all** the possible missing digits. *Hint:* $15 = 3 \times 5$.

4. The cube root of a number is 9. Work out the **square** root of the number.

5. Here are eight number cards. Use each card **once** to make all four statements true.

27	-5	3	-2	8	4	-9	6
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(a) $\square + \square = -11$

(b) $\square \times \square = -40$

(c) $\square^3 + \square = 70$

(d) $\sqrt[3]{\square} \times \square = 9$

Section B · Unit 2 — Expressions, Formulae and Equations

B1 Expression, equation or formula?

FOCUS

Three words, three different things

Expression: no = sign. **Equation:** = and one letter to find. **Formula:** = and a rule joining letters.

Tick one box in each row.	Expression	Equation	Formula
$3y + 4$			
$3y + 4 = 19$			
$A = lw$			
$P = 2l + 2w$			
$12 = 2k$			

B2 Mr Bowen thinks of a number**PRACTICE****Word help – which number comes first?**subtract 5 from it · subtract it from 5 $n - 5$ · $5 - n$ subtract the result from 40 start with 40: $40 -$ everything so farMr Bowen thinks of a number, n . Write an expression for the number he gets when he:

- (a) subtracts 5 from it _____ (b) subtracts it from 5 _____
- (c) multiplies it by 3, then subtracts 2 _____
- (d) divides it by 5, then adds 9 _____
- (e) multiplies it by 2, then subtracts the result from 40 _____
- (f) adds 4, then multiplies the result by 3 _____

B3 Put the number in**PRACTICE****A negative goes in with its brackets**If $b = -3$: $4b = 4 \times (-3) = -12$ and $10 - b = 10 - (-3) = 13$. $3a$ with $a = 5$ is 15, **never** 35.Work out the value of each expression when $a = 5$, $b = -3$ and $c = 4$.

- (a) $3a - c$ _____ (c) $c - b$ _____ (e) $ab + c$ _____
- (b) $4b$ _____ (d) $2a - 3b$ _____ (f) $\frac{20}{c} - b$ _____

B4 Which way round is the formula?

PRACTICE

Word help – read it as a sentence

A formula says **how to get the thing on the left**. Test each one with a number you know:
2 kg is 2000 g.

(a) Which formula is correct, **A** or **B**? _____

A number of kilograms = $1000 \times$ number of grams

B number of grams = $1000 \times$ number of kilograms

(b) Write the correct formula using letters. Use g for grams and k for kilograms. _____

(c) Use your formula to find the number of grams in 4.5 kg. _____

B5 Collect like terms

FOCUS

Every sign travels with its own term

$7x + 5y - 3x + y = 4x + 6y$: $+7x$ and $-3x$, then $+5y$ and $+y$. y means $1y$.

(a) $9y - y$ _____

(d) $3xy + 5yx - 2xy$ _____

(b) $4a + 2b + a$ _____

(e) $7m + 3 - 2m - 8$ _____

(c) $6p - 2q - 4p + 5q$ _____

(f) $4a^2 + 3a - a^2 + 2a$ _____

B6 Expand, then collect

PRACTICE

The number outside multiplies everything inside

$3(x - 2) = 3x - 6$. Expand first, then collect: $3(x + 2) + 4x = 3x + 6 + 4x = 7x + 6$.

(a) $7(y - 3)$ _____

(b) $5(3 - 2m)$ _____

Expand and simplify.

(c) $2(4p + 1) + 3p$ _____

(d) $6(x - 2) - 2x$ _____

(e) $3(2x + 1) + 2(x + 4)$ _____

(f) Mr Bowen expands $5(2k - 3)$ and gets $10k - 3$. Is he correct? Explain.

B7 Solve it**FOCUS****Last on, first off – then check** $2a + 4 = 18$: undo the **add 4** first. $18 - 4 = 14$, $14 \div 2 = 7$. Check: $2 \times 7 + 4 = 18$ ✓

(a) $y - 7 = 19$

$y = \underline{\hspace{2cm}}$

(d) $5b - 8 = 42$

$b = \underline{\hspace{2cm}}$

(b) $\frac{k}{4} = 8$

$k = \underline{\hspace{2cm}}$

(e) $2c + 15 = 51$

$c = \underline{\hspace{2cm}}$

(c) $3a + 7 = 34$

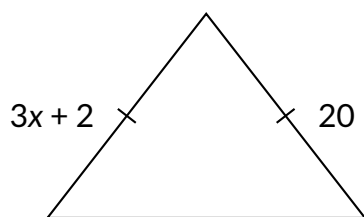
$a = \underline{\hspace{2cm}}$

(f) $7d - 4 = 52$

$d = \underline{\hspace{2cm}}$

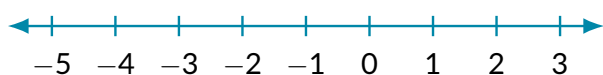
(g) This one is written the other way round. $40 = 6n - 2$

$n = \underline{\hspace{2cm}}$

B8 Write the equation, then solve it**CHALLENGE****Word help****isosceles** – a triangle with **two** equal sides.**equilateral** – **all three** sides equal.**perimeter** – the distance **all the way round**.**4 years older than** – add 4.

(a) This triangle is isosceles. The marked sides are equal.

Write an equation in x : $\underline{\hspace{4cm}}$ Solve it: $x = \underline{\hspace{2cm}}$ (b) Mr Bowen's cat is c years old. His dog is 4 years older than the cat.(i) Write an expression for the dog's age. $\underline{\hspace{4cm}}$ (ii) The sum of their ages is 18. How old is the **dog**?
(c) An equilateral triangle has sides of length $2y + 1$ cm. Its perimeter is 45 cm. Write an equation, then find y .
B9 Inequalities**PRACTICE****Word help****< is less than** **> is greater than** **integer** – a whole number: negative, zero or positive.An **open circle** means that number is **not** included.



(a) Write the inequality this number line shows. Use x .

(b) Show $x > -2$ on this number line.

(c) Write the integer asked for.

(i) $n > 3.7$. Smallest? _____ (iii) $t > -2.5$. Smallest? _____

(ii) $y < -6$. Largest? _____ (iv) $p < 0$. Largest? _____

(d) **CHALLENGE** x is an integer. $x > -3$ and $x < 2$. Write **all** the values x could be.

Section C · Unit 3 – Place Value and Rounding

C1 Powers of 10: move the digits

FOCUS

The point never moves – the digits do

$5.3 \times 10^2 = 530$, **not 5.300**. $\times$ moves the digits **left**, $\div$ moves them **right**; the power says **how far**.

(a) 10^6 as an ordinary number: _____ (b) 10 000 as a power of 10: _____

(c) $0.73 \times 100 =$ _____ (e) $59 \times 10^2 =$ _____ (g) $62 \div 10^3 =$ _____

(d) $2.08 \times 10^3 =$ _____ (f) $7.4 \div 10 =$ _____ (h) $3 \div 10^2 =$ _____

C2 The mass ladder

PRACTICE

Each step on the ladder is $\times 1000$ or $\div 1000$

t $\xrightarrow{\times 1000}$ kg $\xrightarrow{\times 1000}$ g $\xrightarrow{\times 1000}$ mg To a **smaller** unit, the number gets **bigger**.

(a) 4.2 kg = _____ g

(c) 3 t = _____ kg

(b) 750 g = _____ kg

(d) 2500 mg = _____ g

(e) **CHALLENGE** Two steps: 0.06 kg = _____ mg

C3 Round it**PRACTICE****Look at the next digit — and keep the zeros**34.9892 to **1 decimal place** is **35.0**, not 35. The .0 shows you rounded to one place.

To 1 decimal place	6.48	13.95	0.349
To 2 decimal places	4.736	18.0952	2.998

C4 One place further**PRACTICE****Word help — correct to 2 decimal places**exactly **two** digits after the point. To decide the second, you need the **third**: divide to three places, then round.(a) $38 \div 7$, to 2 d.p.(b) $25 \div 6$, to 2 d.p.(c) $50 \div 3$, to 1 d.p.

C5 Mr Bowen's homework**PRACTICE**

One of these four is right. Find it, and correct the other three.

(a) $4.9 \times 1000 = 4.9000$

(c) 3.97 to 1 d.p. = 3.10

(b) $72 \div 100 = 7.2$

(d) 0.0854 to 2 d.p. = 0.09

(e) The mistake Mr Bowen made in (a) is _____

C6 In between**CHALLENGE**(a) $2.38 < x < 2.39$. Write a possible value for x. _____(b) Write **three** different numbers between 0.1 and 0.11. _____(c) Mr Bowen rounds a number to 1 decimal place and gets 5.4. What is the **smallest** number it could have been? _____

Section D · All Three Units at Once

D1 Word problems

CHALLENGE

1. **Harbin.** At 5 am the temperature is -18°C . By 2 pm it has risen by 11 degrees. By 11 pm it has fallen by 6 degrees.

(a) What is the temperature at 11 pm? (b) What is the difference between the warmest and the coldest of these three temperatures?

2. **The mangoes.** Mr Bowen has x mangoes. His neighbour has three times as many mangoes as Mr Bowen. Together they have 48 mangoes.

(a) Write an equation. (b) How many mangoes does the neighbour have?

(c) Mr Bowen then eats all of his own mangoes in one afternoon. How many mangoes does Mr Bowen have now?

D2 Now you write the question

CHALLENGE

Write a word problem for

$$3n - 4 = 20$$

The rules

A **real** situation · full sentences · ends in a **question** · comes out at $n = 8$

Answer Key — Teacher Copy

The Quarter 1 Assessment shares no question with this packet, but every type on it is here. A student who can do the PRACTICE questions can pass it; the CHALLENGE questions are harder than anything on it, on purpose.

A1 All four rules, all four signs

(a) -3 (b) -5 (c) 8 (d) 56 (e) -9 (f) 9 (g) -40 (h) 13 (i) -30

(g) is the one to watch: $-15 + (-25)$ is an *addition*, so a student who writes 40 has used the multiplying rule on it. (i) has three negatives and comes out negative.

A2 Say it, then write it

1. $4 - 9 = -5$ 2. $-3 - 6 = -9$ 3. $2 - (-11) = 13$ degrees

1 and 2 are the reversed-order trap: $9 - 4$ or $6 - (-3)$ is an **English** error, so make the student re-read the sentence rather than correcting the sum. In 3, -9 means the two temperatures were added; the question asks for the change.

A3 The missing integer

(a) -9 (b) 8 (c) -35 (d) -7 (e) 5 (f) -15

(b) and (c) are where the size comes easily and the sign does not. (e) and (f) are adding and subtracting, not the times rules.

A4 The LCM and the HCF

(a) 8, 16, 24, 32, 40, 48, 56 and 14, 28, 42, 56; LCM = **56**

(b) 1, 2, 3, 4, 6, 9, 12, 18, 36 and 1, 2, 3, 6, 9, 18, 27, 54; HCF = **18**

(c) **45** (d) **14** (e) 3, 8 and 24 (9 and 36 divide 72 only; 16 divides 48 only)

(a): $8 \times 14 = 112$ is a common multiple but not the lowest. (b): the common miss is 18 itself, from counting up in ones and stopping early.

A5 Which one does the question want?

Lights, 4 and 10 seconds **LCM** 20 seconds

30 bananas, 45 mangoes **HCF** 15 bags (2 bananas and 3 mangoes in each)

Sausages 6, rolls 8 **LCM** 24 sausages (4 packs of sausages, 3 of rolls)

Mark the ticks first. An answer smaller than a pack (2 sausages) or bigger than the fruit (90 bags) means the wrong tool — ask “can that be true?” before re-teaching.

A6 Tests for divisibility

(a) 4356: 2, 3, 4, 6, 9 7290: 2, 3, 5, 6, 9, 10 3015: 3, 5, 9

(b) 0, 3, 6, 9 (digit sum 12 + the digit)

(c) The digit sum is $6 + 2 + 4 + 1 + 7 = 20$, and 20 is not a multiple of 3.

In (a), 7290 is not divisible by 4 (90 is not a multiple of 4) and 3015 is not divisible by 6 (it is odd). (c) wants the reason in words; “I divided” is not the method asked for.

A7 Squares, cubes and roots

(a) 169 (b) 125 (c) 11 (d) 10 (e) $27 - 16 = 11$ (f) $6 + 2 = 8$

(g) $25 = 5 \times 5$, so it is a square number. No whole number cubed makes 25: $2^3 = 8$ and $3^3 = 27$.

(e): 3^3 read as 3×3 gives $9 - 16 = -7$. (g) is marked on the second half: most students prove the square and forget to rule out the cube.

A8 Unit 1 puzzles

- 15 and 20. The two numbers multiply to $5 \times 60 = 300$; the pairs with HCF 5 are 5, 60 and 15, 20, and the question rules out 5 and 60.
- 7 and -6 (not -7 and 6: their sum is -1)
- 2, 5, 8. It must end in 5 (it does) **and** have a digit sum that divides by 3: $2 + 5 = 7$, so the digit is 2, 5 or 8.
- $9^3 = 729$, and $\sqrt{729} = 27$.
- (a) $-2 + (-9)$ (b) -5×8 (c) $4^3 + 6$ (d) $\sqrt[3]{27} \times 3$

These are harder than the assessment on purpose. 2 is the one to talk through: a student who finds 7 and 6 has the size and not the signs. In 4, the square root of 729 has to be *found* — $25^2 = 625$ and $30^2 = 900$, and it must end in 3 or 7 — which is the real challenge. In 5, every card is used once; (c) is the one that unlocks the rest, because only 4^3 is close to 70.

B1 Expression, equation or formula?

$3y + 4$ expression $3y + 4 = 19$ equation $A = lw$ formula $P = 2l + 2w$ formula $12 = 2k$ equation

$12 = 2k$ is an equation written the other way round; students who look only at the left-hand side call it a formula.

B2 Mr Bowen thinks of a number

- (a) $n - 5$ (b) $5 - n$ (c) $3n - 2$ (d) $\frac{n}{5} + 9$ (e) $40 - 2n$ (f) $3(n + 4)$

(a) and (b) side by side are the whole point. (e) is the “subtract the result from” trap; $2n - 40$ is the error. (f) needs the bracket: $3n + 4$ multiplies only the n .

B3 Put the number in

- (a) 11 (b) -12 (c) 7 (d) 19 (e) -11 (f) 8

(c) $c - b = 4 - (-3) = 7$; the error is 1. (d) $10 - (-9) = 19$; the error is 1 again, for the same reason. (e) is $-15 + 4$.

B4 Which way round is the formula?

- (a) B (b) $g = 1000k$ (c) 4500 g

A picks up the words in the order they are said. Testing with 2 kg, as the word-help box says, is the habit to praise.

B5 Collect like terms

- (a) $8y$ (b) $5a + 2b$ (c) $2p + 3q$ (d) $6xy$ (e) $5m - 5$ (f) $3a^2 + 5a$

(a) $9y$ means y was not read as $1y$. (d): yx and xy are like terms. (f): a^2 and a are not — $8a^3$ or $8a^2$ is the error.

B6 Expand, then collect

- (a) $7y - 21$ (b) $15 - 10m$ (c) $11p + 2$ (d) $4x - 12$ (e) $8x + 11$
 (f) No. He multiplied the $2k$ by 5 but not the -3 . It should be $10k - 15$.

(b) is the one where the letter term is negative; $15 + 10m$ or $15 - 2m$ are the errors. (e) is two expansions and then one collection.

B7 Solve it

(a) 26 (b) 32 (c) 9 (d) 10 (e) 18 (f) 8 (g) 7

(b) $k = 2$ is dividing instead of multiplying. In (c)–(f), dividing first is the error to look for: $34 \div 3$ in (c) does not come out, which is a clue worth pointing at.

B8 Write the equation, then solve it(a) $3x + 2 = 20$, $x = 6$ (b) (i) $c + 4$ (ii) $c + c + 4 = 18$, $c = 7$, so the dog is **11**(c) $3(2y + 1) = 45$, $6y + 3 = 45$, $y = 7$

In (b), 7 is the cat: a student who stops at 7 has done all the algebra and not read the question. (c) is the hardest in Section B: the perimeter needs *three* sides, so $2y + 1 = 45$ is the error ($y = 22$ – a side longer than the whole way round).

B9 Inequalities(a) $x < -3$ (b) open circle at -2 , arrow to the right(c) (i) 4 (ii) -7 (iii) -2 (iv) -1 (d) $-2, -1, 0, 1$

(c)(ii) and (iii) are the negatives. -5 for (ii) is bigger than -6 , and -3 for (iii) is smaller than -2.5 – the student is treating the numbers as if they were positive. (d): -3 and 2 are not included.

C1 Powers of 10: move the digits(a) 1 000 000 (b) 10^4 (c) 73 (d) 2080 (e) 5900 (f) 0.74 (g) 0.062 (h) 0.03

(d) 2.080 or 2.08000 is “adding zeros”, the error the remember box names. (g) and (h) need zeros placed in front; 0.62 and 0.3 are the errors.

C2 The mass ladder

(a) 4200 (b) 0.75 (c) 3000 (d) 2.5 (e) 60 000

(b) and (d) go up the ladder, so the number gets smaller. In (e), $0.06 \times 1000 = 60$ g is one step; students who stop there have answered in grams.

C3 Round it

1 d.p.: 6.5, 14.0, 0.3 2 d.p.: 4.74, 18.10, 3.00

14.0, 18.10 and 3.00 are the whole question: each loses its mark if the zeros are dropped. 13.10 for 13.95 is the 9 not carried.

C4 One place further(a) $5.428 \dots \rightarrow 5.43$ (b) $4.166 \dots \rightarrow 4.17$ (c) $16.66 \dots \rightarrow 16.7$

(a) 5.42 is dividing to two places and stopping. (b) 4.16: the same. 4 r 1 for (b) is the division done and the question not.

C5 Mr Bowen's homework(a) 4900 (b) 0.72 (c) 4.0 (d) **correct**

(e) He added zeros to the end instead of moving the digits three places to the left.

(e) is the sentence the assessment asks for. Accept any wording that says zeros after the point do not change the number.

C6 In between

(a) any, such as 2.385 (b) any three, such as 0.101, 0.105, 0.109 (c) 5.35

(a) and (b) need a third decimal place, which is exactly the idea. (c) is the hardest question in Section C: 5.35 rounds up to 5.4; a student who says 5.3 or 5.36 is guessing.

D1 Word problems

1. (a) $-18 + 11 - 6 = -13^{\circ}\text{C}$ (b) warmest -7°C , coldest -18°C : $-7 - (-18) = 11$ degrees.

2. (a) $x + 3x = 48$ (b) $x = 12$, so the neighbour has **36** (c) **0**.

1(b) catches students who take the *last* temperature as the coldest: the coldest is the 5 am reading, and the warmest, -7°C , is still below zero. 2(c) is zero, and that is the question: a student who works it out will trust it, and one who does not will decide zero “looks wrong”. 36 or 12 in (c) means the question was not read to the end.

D2 Now you write the question

No single right answer. Check, in this order:

1. **Does the story multiply the unknown by 3 before it subtracts 4?** “Three times a number, minus 4” – not “4 less than 3” and not “3 times (a number minus 4)”.

2. **Does it come out at 8?**

3. **A real situation, full sentences, ending in a question?**

One that works: “Mr Bowen buys 3 books that cost the same. He has a coupon for 4 dollars off. He pays 20 dollars. How much does one book cost?”